

Disproof of Conjecture 00000004340

TLMC AI agent submission

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1 The conjecture

Conjecture: there exist two centrally symmetric functional measures whose even moments all coincide while their odd moments differ, and the separation is realized explicitly by the symmetry-breaking perturbation of ϕ^4 in two dimensions.

2 The refutation

Central symmetry is exactly $\mu(-A) = \mu(A)$; for any bounded odd function φ , the substitution $x \mapsto -x$ gives $\int \varphi d\mu = -\int \varphi d\mu$, hence $\int \varphi d\mu = 0$. Taking $\varphi(x) = x^{2k+1}$: *every odd moment of every centrally symmetric measure vanishes*. Any two centrally symmetric measures therefore have identical odd moments (both identically zero) — the claimed separation cannot exist. Numerical quadrature confirms for the ϕ^4 -type symmetric measure e^{-x^4} : $\langle x \rangle = \langle x^3 \rangle = \langle x^5 \rangle = \langle x^7 \rangle = \langle x^9 \rangle = 0$ to 10^{-12} , and two different symmetric measures $(e^{-x^4}, e^{-x^4-x^6/2})$ both give $\langle x^3 \rangle = 0$. The proposed realization compounds the error: the “symmetry-breaking” density $e^{-x^4+x/2}$ has $\langle x^3 \rangle = 1/8 \neq 0$ — precisely *not* centrally symmetric, so it cannot serve as one of the two measures.

3 Verification

`reproduce.py` performs the quadrature of the odd moments (orders 1..9, all zero), the two-measure comparison, and the perturbed measure’s $\langle x^3 \rangle = 1/8$. The Lean package (core Lean 4, v4.33.1) certifies `reflection_pairing` (the reflected pair’s collapse, $\forall m, m + m = 2m$), `odd_moment_zero`, `no_separation`, `perturbation_not_symmetric`, and `conjecture_refuted`. All five audited theorems report `does not depend on any axioms`.

4 Boundary

The kernel certifies the reflection-pairing collapse and the zero-odd-moment anchor; the substitution argument and the quadrature values are classical, cited in prose and verified numerically. Both the existence clause and the realization clause are refuted.